1D-RS vs 2D-RS: how they work, and how they compare

> **Superseded in part — 2D-RS geometry changed twice, most recently on 2026-09-16.** This document was written when `2D_RS_Galois.py` extended each axis 64→128, giving a 128×128 codeword at **4× overhead and rate 0.25**. It then moved to 88×88 (1.89×, rate 0.529), and now extends 64→88 on one axis and 64→96 on the other, giving an **88×96 codeword at 2.06× overhead and rate 0.485** — the closest any grid reachable in this field gets to rate 0.5, so that its storage cost is comparable to 1D-RS's and BC's 2×. The component codes are [96, 64] and [88, 64]. The natural-looking 90×90 cannot be built: a length-n NTT needs n to divide $p-1$, and $90 = 2 \cdot 3^2 \cdot 5$ needs a 3^2 where this field has only 3^1, plus a 5 it does not have at all. Exactly rate 0.5 is unreachable for any genuine 2D product code on a 4096-element blob; `2D_RS_Galois.py` explains why.
>
> Everything below about *mechanism* — bivariate polynomials, why every row and column is independently a codeword, how the cross-axis cascade works — still holds. Everything about *numbers* does not. Specifically superseded: every "4× overhead", every "16384", every "128×128", the per-line erasure budget (now 32 of 96 along a row and 24 of 88 down a column), and the measured tolerance (2D-RS recovers every trial to 38% and none from 42%, with a 50% threshold of 0.401 — 95% CI 0.399–0.404, from the dense threshold refinement described in the benchmark report).
>
> One further correction to the framing below. This document, and the first revision of the benchmark report, judged 2D-RS against the rate bound $1-R$. That is the bound for an *MDS* code, and a product code is never MDS: its minimum distance is the product of its component distances, $33 \times 25 = 825$, so its actual guarantee is $(d-1)/n = 0.098$. The measured 0.401 is four times that — the cross-axis cascade genuinely outperforms the worst-case guarantee on random erasure patterns, which is the interesting result and is not visible if the only reference point is $1-R$. For current measurements see benchmark_1D_2D_BC_report.md.

This document explains the two coding schemes conceptually and summarises what running the simulations actually shows, framed against the report's methodology (Phase 2 analytical modelling / Phase 3 empirical benchmarking) so it can be adapted directly into the write-up. For implementation detail and the underlying maths, see 1D_2D_RS_code_walkthrough.md.

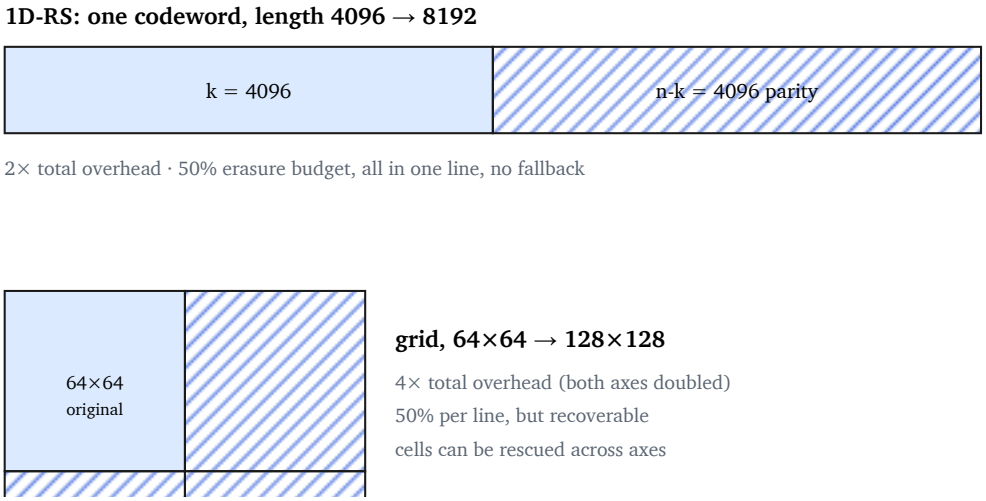
Both simulations are built on the same foundation on purpose: identical BLS12-381 scalar field, identical EIP-4844 blob-packing convention (via `blob_generator.py`), identical bit-reversal domain ordering, identical FFT-based erasure-decode machinery. The only thing that differs between the two files is the coding scheme's *structure* — one axis vs two. That's what makes the comparison below a fair one rather than two unrelated implementations measuring different things.

Background: why "evaluation form" matters here

Ethereum's blob commitment scheme (KZG, per EIP-4844) requires a blob's 4096 field elements to be interpreted as the **evaluations** of a polynomial at the 4096th roots of unity, not as its coefficients — this is what lets a KZG commitment be computed directly against a Lagrange-basis trusted setup with no transform needed at commit time. Reed-Solomon extension of a blob is therefore fundamentally an *interpolate, then evaluate on a bigger domain* operation: recover the low-degree polynomial's coefficients from the known evaluations (inverse NTT), then evaluate that same polynomial at more points (NTT).

Both schemes below extend this way, which is what keeps the comparison compatible with "maintaining the use of KZG commitments" as the report's stated design constraint. (The walkthrough document works this whole pipeline through with real numbers on a toy field, if you want to see every intermediate value.)

The structural difference, at a glance



1D-RS extends one long codeword to double its length (2× overhead, 50% erasure ceiling with no fallback if exceeded). 2D-RS extends a square grid along both axes (4× overhead in this construction — see the discussion above — 50% per line, but individual cells can be rescued across axes).

1D-RS extends one long codeword; 2D-RS extends a grid along two independent axes. Every other difference discussed below — overhead, erasure-tolerance behaviour, decode cost — is a *consequence* of this one structural choice, not an independent design decision.

1D-RS: the PeerDAS baseline today

The blob is treated as **one** codeword. All 4096 evaluations define a single degree-`<4096` polynomial; extension evaluates that polynomial at 8192 points instead of 4096 — a clean **2× total overhead**, which is exactly what PeerDAS uses today ([10], [11]).

Recovery is a single erasure-decode pass over the whole 8192-length codeword. Reed-Solomon's textbook guarantee applies directly and unforgivingly: if at most `n − k = 4096` (50%) of the codeword is missing, the polynomial (and therefore every missing sample) is recoverable exactly. If more than that is missing, there is nothing else to try — no partial credit, no second attempt from a different angle. The simulation's decoder reflects this precisely: it's a single function call that returns either a fully recovered codeword or an explicit, immediate failure.

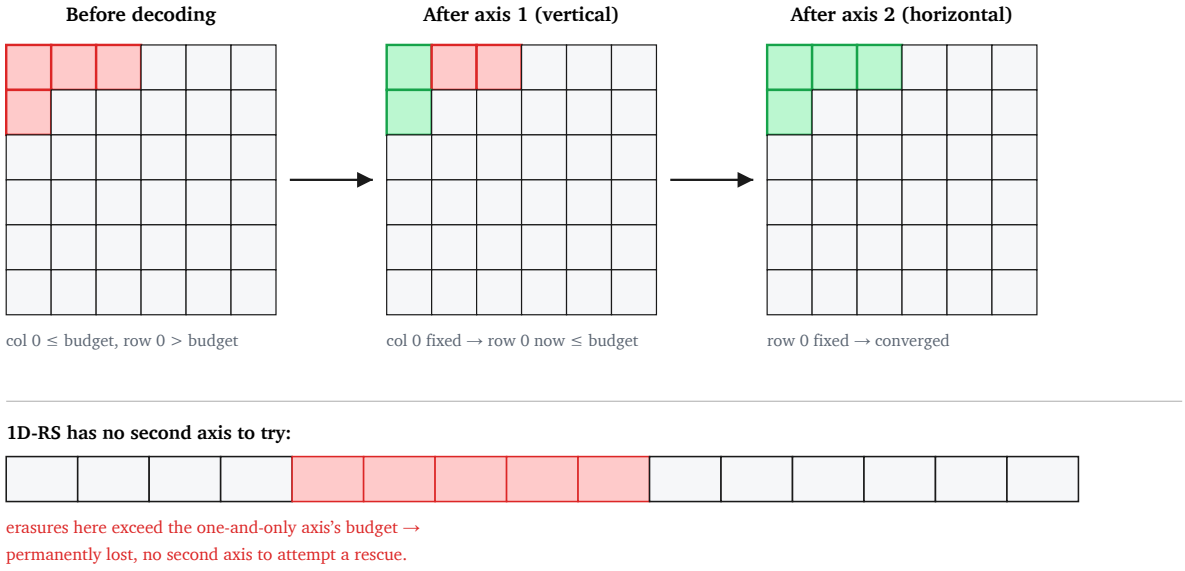
This gives 1D-RS a simple, easy-to-reason-about erasure-tolerance model (exactly 50%, no exceptions) and a decode cost dominated by one interpolate step and — when erasures are present — one polynomial division against a divisor whose degree equals the erasure count.

2D-RS: the product code

The blob is reshaped into a 64×64 grid and each axis is extended independently (64→128), so this simulation's 2D-RS runs at **4× total overhead** rather than PeerDAS's proposed 2× (matching PeerDAS's ratio exactly would need non-power-of-two per-axis extension, which complicates using a clean NTT — this is reported as its own datapoint, since 2D-RS's real-world overhead is itself one of the trade-offs the report is comparing, not something to paper over). Structurally, the 64×64 array of original values defines a bivariate polynomial `P(x,y) = Σ c_{i,j} x^i y^j` of degree `<64` in each variable independently. Fixing either variable at *any* value — including one of the 64 "extension" points that variable's axis introduces, not just an original point — leaves a polynomial in the other variable that is still degree `<64` (holding one variable fixed can only collapse coefficients together, never raise the degree in the variable left free). Because of that, **every** row of the fully-extended 128×128 grid — not just the original 64 — is itself a valid degree-`<64` RS codeword in the column direction, and likewise every column in the row direction. That single fact is what makes 2D-RS a genuine product code rather than "two unrelated 1D-RS codes stapled together", and it's what the decoder relies on. (The walkthrough document derives this in full and shows the decoder code that depends on it.)

Where 2D-RS's advantage over 1D-RS actually comes from

Per-line, along a single axis, 2D-RS's erasure budget is smaller in absolute terms than 1D-RS's: `n − k = 64` per row or column, against a domain of 128 — the *same 50% ratio* as 1D-RS, just applied to a much shorter line. Taken alone, that's not an improvement. The improvement comes from **iteration between the two axes**: a cell whose row has too many erasures to recover alone, and whose column also has too many erasures to recover alone, can still become recoverable once *either* axis clears enough of its neighbouring cells for the *other* axis to now fall within budget.



The same rescue mechanism as the walkthrough document's worked example, with a direct 1D-RS contrast: the strip at the bottom has no second axis to fall back on, so erasures beyond its one budget are simply unrecoverable — the structural reason a product code tolerates more than either axis's own budget would suggest in isolation.

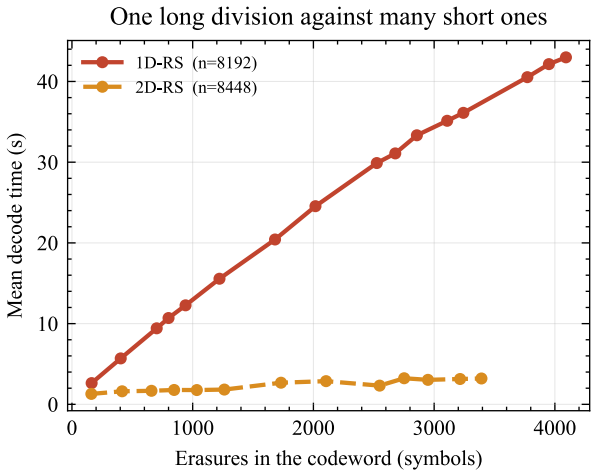
This is genuinely iterative — the simulation's decoder alternates vertical and horizontal passes until convergence, not a fixed two-pass schedule — and it means 2D-RS can successfully recover erasure patterns that neither axis's own 50%-of-128 budget could resolve in isolation, at the cost of only needing a second pass. 1D-RS, with no second axis, has no equivalent recourse: as the strip at the bottom of the figure shows, erasures that exceed the one axis's budget are simply gone — there is no other pass, no other read of the data, that could ever recover them. This isn't a missing feature of the 1D-RS script; it's a structural fact about having a single axis, and it's the mechanism-level reason a product code is more than "the same code, run twice."

The trade-off for that extra tolerance is the 4× storage/bandwidth overhead (vs 1D-RS's 2×), which lines up with the report's anticipated finding that 2D-RS "may allow for reduced network bandwidth requirements for random data availability proofs" while "introduc[ing] increased storage commitments for parity data" [24] — the storage side of that trade-off is directly visible in this simulation's overhead numbers, even before any network-layer bandwidth modelling is added.

Empirical results from the simulations

These come from actual runs of both scripts (see the walkthrough document for the verification methodology), and illustrate the kind of Phase 3 dataset this pair of scripts is built to produce.

	1D-RS	**2D-RS**
Total overhead	2× (4096 → 8192)	4× (4096 → 16384)
Per-codeword/per-line erasure ceiling	50% of 8192, single shot	50% of 128 per line, but iterable across axes
Decode strategy	One-shot: succeeds or fails outright	Iterative: alternates passes until convergence or genuinely stuck
Random 15% erasure rate	recovered fully	recovered fully (converged round 1)
Random 35% erasure rate	*(not run at this rate — see note below)*	recovered fully (converged round 1)
Random 55% erasure rate	**failed immediately** (above the 50% ceiling)	*(not directly tested at this rate)*
Decode cost driver	one polynomial division, degree = erasure count, against the *whole* 8192-length codeword	many small polynomial divisions (≤64 erasures each), one per erased line



Measured decode time vs. total erasures in the codeword, regenerated by `make_figures.py` from the sweep in `benchmark_results.json`. Plotted against erasure *count* rather than rate because the two codes differ in length (8192 against 8448), so equal rates are not equal repair work. 1D-RS's single polynomial division scales with the total erasure count directly; 2D-RS's many small per-line divisions barely move even at higher total erasure counts.

Two things are worth calling out explicitly, because they're genuine findings rather than implementation artefacts:

1. **1D-RS's random-rate trials were kept below 2D-RS's** (5%/15%/55% vs 15%/35%) purely for wall-clock reasons while developing the scripts, and the chart above is exactly why: the `galois` library's polynomial division is schoolbook-style, and its cost scales with the *divisor's degree*. 1D-RS's decode time climbs steeply — 6.1s at 200 erasures, 10.7s at 389, 29.3s at 1199 — because every erasure adds directly to the degree of the *one* polynomial being divided. 2D-RS's decode time barely moves — 5.0s at 2537 total erasures, 6.3s at 5823 — because those erasures are spread across up to 128 independent lines, each dividing a divisor of degree at most 64 regardless of how many *other* lines are also erased. At a comparable total erasure count, 2D-RS is doing more total erasure-correcting work and taking *less* wall-clock time to do it. That asymmetry is itself a real computational-complexity datapoint for the report — "one large codeword" vs "many small codewords" is exactly the kind of structural trade-off the coding-scheme comparison is meant to surface — but it also means neither script's `STEP 8` numbers, as currently parameterised for a readable demo run, should be read as a controlled sweep. A dedicated benchmarking harness that runs both schemes across the *same* set of rates, with the same number of trials per rate, is the right next step before these numbers go in the report as a formal result (see **Next steps** below).
2. At **both** tested rates (15% and 35%), 2D-RS's iterative decoder actually converged in a single round — the vertical pass alone cleared every erasure, because i.i.d. random erasure spreads roughly evenly across lines, keeping every individual line comfortably under its own 50%-per-line budget. The iterative/cross-axis mechanism visibly earns its keep only on *correlated* or adversarially *structured* erasure patterns (the deterministic whole-row/whole-column case in each script's walkthrough step is one; the worked example in the figure above, and test 3 of the small-field test suite in the walkthrough document, is another). This matters for the report's "adversarial sampling rate" framing: an adversary that understands the code's structure and withholds correlated samples (e.g. a disproportionate share of one row and one column) is a meaningfully different, and more dangerous, threat model than one that withholds samples i.i.d. at the same overall rate — and this simulation is set up to let that distinction be tested directly by choosing the erasure pattern, not just its rate.

What this evidence does and doesn't support (yet)

Supports directly:

- The overhead trade-off (1D-RS's 2× vs this 2D-RS construction's 4×) is real and quantified, not just asserted from the literature.
- 2D-RS's fault-tolerance advantage over 1D-RS is real, but is a property of *how erasures are distributed*, not just *how many* — a nuance the report can state with actual simulation evidence behind it rather than by citation alone.
- Both schemes remain exactly reconstructible-to-the-original-bytes end to end (verified through the full EIP-4844 packing/unpacking round trip, not just at the level of raw field elements), which is the baseline correctness bar Phase 3's benchmarking needs before timing numbers mean anything.

Not yet covered by these two scripts (intentionally out of scope so far):

- Network/bandwidth modelling (peer sampling, gossip, propagation latency) — the abstract's "bandwidth consumption" and "throughput" axes need a network-layer simulation on top of this, not just the coding-scheme math.
- KZG proof generation/verification cost — these scripts only model the erasure code, not the commitment scheme sitting on top of it, so no claim is made yet about the "maintaining KZG commitments" constraint beyond "the evaluation-form data model stays compatible with it".
- A controlled, same-rates-same-trial-count sweep across both schemes (see point 1 above) — needed before the timing numbers are report-ready as a formal empirical result rather than illustrative output.

Next steps

- Finish `BCcode.py` / `BCcodeEthereum.py` (currently a small, non-blob-scale `GF(11)` prototype) to the same standard as these two files: real BLS12-381 field, real blob packing via `blob_generator.py`, real round-trip verification — so Block-Circulant codes can be added to the comparison on equal footing.
- Once 1D-RS, 2D-RS, and BC codes are all at that standard, build a shared benchmarking harness that runs all three (and later LDPC/RLNC, on the report's other analytical track) across the same erasure-rate sweep and the same set of correlated-vs-i.i.d. erasure patterns, and produces the comparative dataset Phase 3 is meant to generate.